Q1.

John Newlands was a chemist who worked in a sugar factory.

In 1866 he designed a periodic table.
He arranged the elements in order of their relative atomic masses.

He found a repeating pattern for some of the elements.
Newlands wrote, 'the eighth element starting from a given one, is a kind of repetition of the first, like the eighth note in an octave of music'.

H	**Li**	**G**	**Bo**	**C**	**N**	**O**
F	**Na**	**Mg**	**Al**	**Si**	**P**	**S**
Cl	**K**	**Ca**	**Cr**	**Ti**	**Mn**	**Fe**
Co, Ni	**Cu**	**Zn**	**Y**	**In**	**As**	**Se**
Br	**Rb**	**Sr**	**Ce, La**	**Zr**	**Di, Mo**	**Ro, Ru**
Pd	**Ag**	**Cd**	**U**	**Sn**	**Sb**	**Te**
I	**Cs**	**Ba, V**	**Ta**	**W**	**Nb**	**Au**
Pt, Ir	**Tl**	**Pb**	**Th**	**Hg**	**Bi**	**Os**

Newlands' periodic table

(a) In Newlands' periodic table, the elements lithium, sodium and potassium are grouped together.

Give **two** properties of these elements which support the idea that they should be grouped together.

1 .....

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2 .....

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(2)

(b) Newlands' periodic table was not accepted by most chemists in 1866.

Suggest reasons why.
Use the Newlands' periodic table above to help you to answer this question.

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(3)

(c) State **and** explain **one** way in which Mendeleev improved Newlands' periodic table.

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(2)

(Total 7 marks)

Q1.

(a) any **two** from:

- react with water **or** very reactive
- (react with water) releasing gas / hydrogen / fizzing
- (react with water) to form an alkaline / hydroxide solution
- form ions with a 1+ charge
  allow lose one electron from the outer shell
  ignore other references to electronic structure
  ignore physical properties

2

(b) any **three** from:

- some boxes contain two elements
  allow specific examples:
  *Co, Ni **or** Ce, La **or** Di, Mo **or** Ro, Ru **or** Ba, V **or** Pt, Ir*
- groups / columns contain elements with different properties
  allow groups / columns contain both metals and non-metals
  ignore examples
- Newlands not a well-known / respected scientist
  ignore references to sugar factory
- new idea (not readily accepted by other scientists)
  allow musical scales thought to be silly by some scientists

3

(c) one for improvement **and** one for explanation from:

- left gaps (for undiscovered elements) (1)
- so that elements were in their correct group (1)
  allow so the elements fitted the pattern of properties

  or

- did not always follow order of relative atomic weights / masses (1)
  ignore references to atomic number / electronic structure
- so that elements were in their correct group (1)
  allow so the elements fitted the pattern of properties

2

[7]